2017/2018 Exam Paper Section B

Q2a) ii) cont.

$$= \pi_K = \frac{p_1}{p_2} \pi_{K-1} = \rho \pi_{K-1}$$

so (1) holds for K

$$\Rightarrow \pi_K = \rho^K \pi_0$$

We know $\sum_{K=0}^{r} \pi_K = 1$

$$\Rightarrow \sum_{K=0}^{r} \rho^K \pi_0 = 1$$

$$\Rightarrow \pi_0 = \frac{1}{\sum_{K=0}^{r} \rho^K}$$

$$\Rightarrow \pi_0 = \frac{1}{\frac{1-\rho^{r+1}}{1-\rho}} = \frac{1-\rho}{1-\rho^{r+1}}, \ \rho \neq 1$$

$$\pi_0 = \frac{1}{r+1}, \ \rho = 1$$

Q2b) $\alpha_K = \alpha$, $\beta_K = K\beta$, $\alpha > 0$, $\beta > 0$

Show π_K, $K = 0, 1, 2, \ldots, r$ are given by

$$\pi_K = \frac{\rho^K}{K!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \ldots + \frac{\rho^r}{r!}} \quad \text{where } \rho = \frac{\alpha}{\beta}$$

$$\pi_K = \frac{\rho^K}{K!} \pi_0, \quad \pi_K = \frac{\rho^K}{K!} e^{-\rho}$$

Since $\sum_{K=0}^{r} \pi_K = 1$ we get $\pi_0 \sum_{K=0}^{r} \frac{\rho^K}{K!} = 1$

$$\pi_0 = \frac{1}{\sum_{K=0}^{r} \frac{\rho^K}{K!}}$$

$$\Rightarrow \pi_K = \frac{\rho^K}{K!} e^{-\rho} \frac{1}{\sum_{i=0}^{r} \frac{\rho^K}{i!}} \quad K = 0, 1, \ldots, r.$$

2017/2018 Exam Paper Section B

Q2 b) cont.

Then $\pi_K = \frac{\rho^K}{K!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \dots + \frac{\rho^r}{r!}}$

Q2 c)

phone exchange with 3 operators

30 people call the exchange per hour

each call is serviced in 4 minutes

arrivals = α, servicings = β

$$\alpha = \frac{30}{60} \text{ min}^{-1} = \frac{1}{2} \text{ min}^{-1}$$

$$\beta = \frac{1}{4} \text{ min}^{-1}$$

$$\rho = \frac{\alpha}{\beta} = \frac{1/2}{1/4} = 2$$

$r = 3$

$$\pi_3 = \frac{2^3}{3!} \frac{1}{1 + 2 + \frac{2^2}{2!} + \frac{2^3}{3!}}$$

$$\pi_3 = \frac{4}{3}\left(\frac{3}{19}\right)$$

$$\pi_3 = \frac{4}{19}$$

2017/2018 Exam Paper Section B

Q3 a) r is not the growth rate. Provided the $P < r^{-1}$ then $R(P) < 1$ and population declines.

Q3 b) $P_{n+1} = R(P_n) P_n$,

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}$$

Fixed points where $R(P) = 1$

i.e. $P^2 - rK^2 P + K^2 = 0$

$$P_{n+1} = R(P_n) P_n$$

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}$$

if $0 < P \ll K$ then $R(P) = rP$

$$\Rightarrow P_{n+1} = r P_n^2$$

$P^* = 0$ is a trivial fixed point

Let $R(P) = 1 \Rightarrow \dfrac{rP}{1 + (P/K)^2} = 1$

$$rP = 1 + \left(\frac{P}{K}\right)^2$$

$$P^2 - K^2 r P + K^2 = 0$$

two roots $= P^*$ & P_{crit} $\quad P^* > P_{crit}$

$$(P - P^*)(P - P_{crit}) = 0$$

$$R(P) < 1 \Rightarrow \frac{rP}{1 + P^2/K^2} < 1$$

$$\Rightarrow \frac{P^2}{K^2} - rP + 1 > 0$$

$$(P - P^*)(P - P_{crit}) > 0$$

$$P < P_{crit} \Rightarrow R(P) < 1$$

$$P^* = \frac{1}{2} rK^2 \left(1 + \sqrt{1 - \left(\frac{2}{rK}\right)^2} \right)$$

$$P_{crit} = \frac{1}{2} rK^2 \left(1 - \sqrt{1 - \left(\frac{2}{rK}\right)^2} \right)$$

2017/2018 Exam Paper Section B

Q3 c) i) $r = 2$ $\quad k = 10$ $\quad P_0 = 1$ $\qquad R(P) > 1$

$$P^* = \frac{1}{2}(2)(10)^2\left(1 + \sqrt{1 - \left(\frac{2}{(2)(10)}\right)^2}\right) = 199.499$$

$$P_{crit} = \frac{1}{2}(2)(10)^2\left(1 - \sqrt{1 - \left(\frac{2}{(2)(10)}\right)^2}\right) = 0.501$$

$P_0 = 1$

$P > P_{crit}$

$1 > 0.501$

$0.501 < 1 < 199.499$ $\Rightarrow$ population increases and is bounded above by P^*

Q3 c) ii) $P_0 = 50$ $\qquad R(P) > 1$

$P > P_{crit}$

$0.501 < 50 < 199.499$

$\Rightarrow$ population increases & is bounded above by P^*

Q3 c) iii) $P_0 = 100$ $\qquad R(P) > 1$

$P > P_{crit}$

$0.501 < 100 < 199.499$.

$\Rightarrow$ population increases & is bounded above by P^*